

5.4 Brittle Bones

- ▼ What is a special characteristic of the amino acid cysteine?
 - It can form disulphide bridges between two of its residues which can lead to the linkage seen
- ▼ What consequences could result from the change of amino acid residue glycine to cysteine in collagen?
 - Glycine is the smallest amino acid as it can fit down the middle of the collagen triple helix
 - Changing this to cysteine could result in **improper assembly of the triple helix** hence giving the collagen poor functionality as it cannot fold well into fibres.
- ▼ Why does heterozygosity for mutated collagen $\alpha 1$ chain not just result in 50% of alpha chains to be abnormal?
 - It should be like this in an ideal world, but there are so many factors that play a part here
 - There may be **differences in the rates of transcription of the gene, rate of translation, stability of mRNA or stability of the protein** which lead to a different ratio of abnormal and normal chains.
- ▼ How does abnormal collagen type I affect the formation of bone?
 - The bone is formed when **hydroxyapatite** (form of **calcium phosphate**) is **laid down upon a collagen type I scaffold**
 - The **abnormal collagen type I** leads to **defects in the mineralisation process**, which ultimately results in weak bones.
- ▼ What other parts of the body can be affected by abnormal collagen formation?
 - Eyes
 - Teeth
 - Skin
 - Ears

- ▼ Describe a prenatal technique that can be used to detect genetic abnormalities (i.e mutations).
- Chorionic villus sampling or amniocentesis can be used to obtain the DNA
 - PCR can be used to amplify the DNA, and analyse the abnormalities
 - A limitation is that we have to know the existing mutation in order to be able to test for it (i.e adding a probe to see how the DNA interacts with it etc.)
- ▼ What is restriction fragment length polymorphism (RFLP)?
- An RFLP probe is a labeled **DNA sequence that hybridises with one or more fragments of the digested DNA sample** after they were separated by gel electrophoresis, thus revealing a unique blotting pattern characteristic to a specific genotype at a specific locus.

